Kedar Joshi

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EDUCATION

University of Southern California, Los Angeles, CA **Aug 2026 – May 2028**
MS in Computer Science

Manipal University Jaipur, Jaipur, India **Graduated May 2026**
Bachelor of Technology in Computer Science with *Distinction* (CGPA 8.88/10)
Relevant Courses: Data Structure and Algorithms, Artificial Intelligence and Soft Computing, Data Science and Machine Learning

PUBLICATIONS

Don't Judge a Book by its Cover: Testing LLMs' Robustness Under Logical Obfuscation **EACL 2026 (main conference)**
Abhilekh Borah, Shubhra Ghosh*, **Kedar Joshi***, Aditya Kumar Guru*, Kripabandhu Ghosh. **Equal contribution.*

- Introduced Logifus, a structure-preserving logical obfuscation algorithm, and LogiQAte, a 1,108-question diagnostic benchmark across four reasoning tasks.
- Frontier models degrade by 28.8% on average under logically equivalent rewrites, up to −86% in the worst case, with a 50–80% collapse in late-layer next-token confidence.
- Code: github.com/abhilekhborah/logical-obfuscation · Data: huggingface.co/datasets/abhilekhborah/LogiQAte

Contrastive Activation Decomposition: Activation Steering Mitigation by Leveraging Bias Geometry **Preprint, 2026**

- Diagnostic that prescribes intervention rank and target layers before any steered generation, by decomposing contrastive activation differences.
- Bias geometry varies by type: housing discrimination is near rank-1 (EVR_1 = 67.9%), medicalization bias is diffuse (EVR_1 = 10.2%), explaining why rank-1 steering methods fail on one and succeed on the other.
- Matches CAA's debiasing on AccessEval (d = 0.337 vs 0.341) with no per-task tuning, at a fifth of the output-quality cost; validated on Llama, Qwen, Gemma, and Mistral models.

Integrating GLCM Texture Analysis for Improved Deepfake Detection on CelebDF(v2) Dataset **ACOIT 2024, IEEE**
Kedar Joshi, Aditya Sinha. *First author*.

- Texture and frequency features with gradient-boosted trees, detecting manipulation without a CNN forward pass.
- Targets resource-constrained and edge deployment where CNN-based detectors are impractical.
- ieeexplore.ieee.org/document/10939531

PROFESSIONAL EXPERIENCE

Indian Institute of Science Education and Research (IISER), Kolkata, India **May 2025 – Nov 2025**
Research Intern – LLM Evaluation and Robustness

- Evaluated six state-of-the-art models (GPT-4o, GPT-5, o4-mini, Claude, Gemini, Qwen) for reasoning robustness under logical obfuscation; published at EACL 2026 (see Publications).

India Meteorological Department, Bhopal, India **Nov 2024 – Feb 2025**
Research Intern – AI\ML for Atmospheric Sciences

- Filed a patent application for automated physics-informed neural architecture search for single-radar 3D atmospheric velocity field reconstruction.
- 3D wind retrieval has required two or more Doppler radars since the 1960s, because a single radar measures only the radial velocity component. This is the first method to reconstruct the full field from one radar, using conservation laws in place of the geometric constraints the second radar provided.
- Architecture search combining DARTS and Bayesian optimization, treating physics compliance as a third objective alongside accuracy and efficiency, across 245K to 50M+ parameters.

ACADEMIC PROJECTS AND PAPERS

Perturbation-Aware Fine-Tuning (PAFT) for Reducing LLM Memorization

- Training-time data augmentation method reducing memorization in LLMs without sacrificing reasoning performance, with MemGenTOI, a metric for the memorization-generalization trade-off.
- Consistent memorization reduction across Qwen2.5 7B, LLaMA 3.1 8B and Mixtral 8x7B on GSM8K; released GSM8K-P, 5,814 perturbation-validated samples for privacy-safe training.

TECHNICAL SKILLS

- **Programming languages:** Python, C
- **Deep Learning Frameworks:** Pytorch, Tensorflow
- **ML/AI Libraries:** NumPy, Pandas, scikit-learn, OpenCV, scikit-image, Matplotlib, Unsloth
- **Specialized Skills:** Physics-Informed Neural Networks (PINNs), Neural Architecture Search (NAS), Computer Vision, Natural Language Processing, LLM Evaluation, Benchmark Development
- **Tools and Technologies:** CUDA, OpenAI, Prover9, Git

COURSES & CERTIFICATIONS

- **Generative AI with Large Language Models** | DeepLearning.AI & AWS | Feb 2025
- **Prediction and Control with Function Approximation** | University of Alberta | Dec 2024
- **Sample-based Learning Methods** | University of Alberta | Nov 2024
- **Fundamentals of Reinforcement Learning** | University of Alberta | Nov 2024